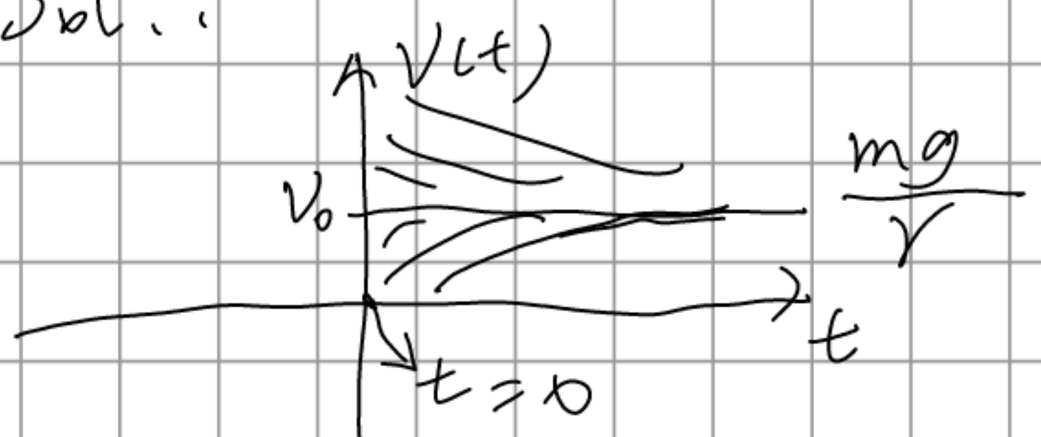
1

Ch1 sec. 1.1~1.3

$$\begin{cases} \dfrac{dv(t)}{dt} = g - \dfrac{\gamma}{m} v(t) \\ v(0) = v_0 \end{cases}$$ ②

Sol.:



$$\frac{1}{g - \frac{\gamma}{m} v(t)} \frac{dv(t)}{dt} = 1$$

$$v(t) = \frac{mg}{\gamma} + \left[v_0 - \frac{mg}{\gamma} \right] e^{-\frac{\gamma}{m} t}$$

Well-posted d.e.'s

① existence

② uniqueness

③ Continuity w.r.t. initial data ①

ill-posted d.e.'s

E.g.:

1. $\begin{cases} \frac{dy}{dt} = y^{\frac{1}{3}} \\ y(0) = 0 \end{cases}$

→ ... → $y_1(t) = 0$ is a sol.

$$\frac{1}{y^{\frac{1}{3}}} \frac{dy}{dt} = 1$$

$$\frac{3}{2} \frac{d}{dt}\left(y^{\frac{2}{3}}(t)\right) = \left(\frac{2}{3}t\right)'$$

$$\Rightarrow y^{\frac{2}{3}} = \frac{2}{3}t + C$$

$$\Rightarrow y(t) = \left(\frac{2}{3}t\right)^{\frac{3}{2}}$$

$$0 = y(0) = \frac{2}{3} \cdot 0 + C = C$$

2. (Non-uniqueness of 1.)

① $\begin{cases} \frac{dy}{dt} = y^2 \\ y(0) = 1 \end{cases}$

$$\frac{1}{y^2} \frac{dy}{dt} = 1$$

$$\Rightarrow \frac{d}{dt}\left(-\frac{1}{y(t)}\right) = 1 = \frac{d}{dt}t$$

$$\Rightarrow -\frac{1}{y(t)} = t + C$$

$$-\frac{1}{y(0)} = 0 + C \Rightarrow C = -1$$

$$\Rightarrow y(t) = \frac{1}{1-t}$$

$t \in [0, 1)$

$\lim_{t \to 1} y(t) = \infty$

② $\begin{cases} \frac{dy}{dt} = y^2 \\ y(2) = 1 \end{cases}$

$$-\frac{1}{y(2)} = 2 + C \Rightarrow C = -1$$

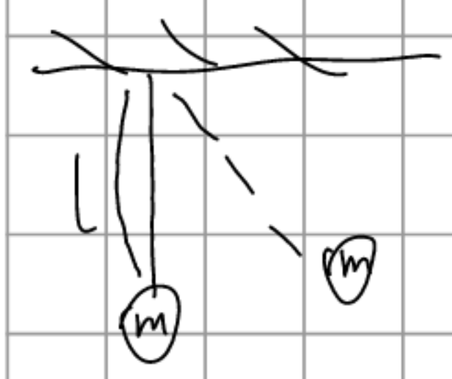
$$-\frac{1}{y(2)} = 1$$

$y(t) =$

$$\rightarrow y = \frac{1}{3-t}$$

②

The pendulum problem



設无friction，
机械能守恒

Formula: $W = F \cdot d$

势能 $E_p = F \cdot d = mgl(1-\cos\theta)$

动能 $E_k = \frac{1}{2} m v^2(t)$

$d = l - l\cos\theta$

$v = \omega \cdot l = \frac{d\theta(t)}{dt} \cdot l$

角速度

$\rightarrow E_k = \frac{1}{2} m \left(\frac{d\theta(t)}{dt}\right)^2 l^2$

$E = E_k + E_p = \cdots$

~~System~~ $\Leftarrow \frac{dE(t)}{dt} = 0$ ← 机械能守恒

$\Rightarrow \frac{dE}{dt} = \frac{1}{2} m l^2 \cdot 2 \frac{d\theta(t)}{dt} \cdot \frac{d^2\theta(t)}{dt^2} + mgl \cdot \sin\theta \cdot \frac{d\theta}{dt} = 0$

$\Rightarrow \frac{d\theta(t)}{dt}\left(l \frac{d^2\theta(t)}{dt^2} + g\sin\theta(t)\right) = 0$

Eq. of motion $\Rightarrow \frac{d^2\theta(t)}{dt^2} + \frac{g}{l}\sin\theta(t) = 0$

Small Oscillations ($|\theta(t)| < 1$): $\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots$

→ 对于小 θ, $\sin\theta \approx \theta$

$\rightarrow \frac{d^2\theta}{dt^2} + \omega^2\theta = 0$ where $\omega^2 = \frac{g}{l}$

Try $\theta(t) = \sin(\omega t)$

$$\theta''(t) = -\omega^2 \sin(\omega t) = -\omega^2 \theta$$

$$\theta'' + \omega^2 \theta = 0$$

$\theta(t) = \cos(\omega t)$ is another solution

$\theta(t) = A\cos(\omega t) + B\sin(\omega t)$

一般

Ch 2 Sec. 2.1

$$\frac{dy(t)}{dt} + p(t)y(t) = g(t)$$

(-4) First, linear diff. eq.

with non-constant but cont. func.'s

→ When $p(t)=0$: $y(t) = \int g(t)\,dt$

Method of integrating factor

$$\mu(t)\frac{dy}{dt} + \mu(t)p(t)y(t) = \mu(t)g(t),$$

这里的 $\mu(t)$ 待定

$$\frac{d}{dt}(\mu(t)y(t)) = \frac{d\mu(t)}{dt}y(t) + \mu(t)\frac{dy(t)}{dt}$$

Demand $\frac{d\mu(t)}{dt} = \mu(t)p(t)$

$$\Rightarrow \frac{1}{\mu}\frac{d\mu}{dt} = p$$

$$\Rightarrow \frac{d}{dt}\ln|\mu(t)| = p(t)$$

$$\Rightarrow \ln|\mu(t)| = \int p(t)\,dt$$

$$|\mu(t)| = e^{\int p(t)\,dt}$$

$$\mu(t) = \pm e^{\int p(t)\,dt}$$

Pick $\mu(t) = e^{\int p(t)\,dt}$

$\frac{dy}{dt} + py = g$ becomes $\frac{d}{dt}(\mu y) = \mu g$

Thus $\mu y = \int \mu(t)\, g(t)\,dt + C$

$$y(t) = \frac{1}{\mu(t)}\left[\int \mu(t)\, g(t)\,dt + C\right]$$

↳ solution

E.g.: $\frac{dy}{dt} + ay = b$, where $a, b \in \mathbb{R}$

Step 1: $\mu(t) = e^{\int a\,dt} = e^{at}$

$$e^{at}\frac{dy}{dt} + a e^{at} y = b e^{at}$$

$$\frac{d}{dt}\left(e^{at} y(t)\right) = b e^{at}$$

$$e^{at} y(t) = \int b e^{at}\,dt = \frac{b}{a} e^{at} + C$$

$$y(t) = C e^{-at} + \frac{b}{a}$$

Initial Value Problem

$$\frac{dy}{dt} + ay = b$$

$$y(0) = 1$$

① $y(t) = ce^{-at} + \frac{b}{a}$

$1 = y(0) = ce^{-a\cdot 0} + \frac{b}{a} = c + \frac{b}{a}$

$\Rightarrow c = 1 - \frac{b}{a}$

$y(t) = (1 - \frac{b}{a})e^{-at} + \frac{b}{a}$

$= e^{-at} + \frac{b}{a}(1 - e^{-at})$

$\rightarrow$ unique solution

② $\begin{cases} \frac{dy}{dt} + ay = g(t) \\ y(t_0) = y_0 \end{cases}$ $\quad \mu(t) = e^{at}$

$y(t) = \frac{1}{\mu(t)}\left[\int_{t_0}^{t} \mu(s)g(s)\,ds + C\right]$

$y_0 = y(t_0) = e^{-at_0}[0 + C]$

$\Rightarrow C = y_0 e^{at_0}$

Unique $y(t) = e^{-at}\left[\int_{t_0}^{t} e^{as}g(s)\,ds + e^{at_0}y_0\right]$

$= y_0 e^{-a(t-t_0)} + \int_{t_0}^{t} e^{-a(t-s)}g(s)\,ds$

↖ Sol. of ··· ↗ Sol. of ···

··· (적분)

3

$$\frac{dy}{dt} + ay = g(t) \qquad p(t) \to 0 \to g(t)$$

$$y(t_0) = y_0$$

$$\mu(t) = e^{\int p(t)\,dt} = e^{at}$$

$$e^{at} y' + a y e^{at} = g(t) e^{at}$$

$$\frac{d}{dt}(e^{at} y) = g(t) e^{at}$$

$$e^{at} y(t) = \int_{t_0}^{t} g(s) e^{as}\,ds + C$$

$$e^{at_0} y_0 = \int_{t_0}^{t_0} 1\,ds + C = C$$

$$\Rightarrow \cdots$$

$$\textcircled{2} \Rightarrow y(t) = e^{-a(t-t_0)} y_0 + \int_{t_0}^{t} e^{-a(t-s)} g(s)\,ds$$

$$= \int_{t_0}^{t} g(s) e^{as}\,ds$$

$$= e^{at}$$

$$= y_n(t) + y_{nh}(t)$$

$y_h(t) = e^{-a(t-t_0)} y_0$ and it is a sol. of $\begin{cases} y' + ay = 0 \\ y(t_0) = y_0 \end{cases}$

$y_{nh}(t) = \int_{t_0}^{t} e^{-a(t-s)} g(s)\,ds$ and it is a sol. of $\begin{cases} y' + ay = t \\ y(t_0) = y_0 \end{cases}$

$\Rightarrow$ Duhamel's Principle: $y(t) = y_h(t) + y_{nh}(t)$

The method of the variation of parameters

$$y' + p(t)\,y = g(t)$$

Step 1: $y' + p(t)\,y = 0$

$$\frac{dy}{dt} = -p(t)\,y$$

$$\frac{1}{y}\frac{dy}{dt} = -p$$

$$\Rightarrow \ln|y(t)| = -\int p(t)\,dt + C$$

$$\frac{d}{dt}\ln|y(t)| = -p(t)$$

$$y(t) = \pm e^{-c}\, e^{-\int p(t)\,dt}$$

$$y(t) = C\, e^{-\int p(t)\,dt}$$

Step 2: Try $y(t) = C(t)\, e^{-\int p(t)dt}$

into

$$\frac{d}{dt}y(t) = C'(t)\, e^{-\int p(t)\,dt} + \underbrace{C(t)\, e^{-\int p(t)dt}\cdot(-p(t))}_{=-p(t)\,y(t)}$$

$$\underbrace{\frac{d}{dt}y + py}_{\Rightarrow g(t)} = C'(t)\, e^{-\int p(t)dt}$$

$$C'(t) = g(t)\, e^{\int p(t)\,dt} = g(t)\,\mu(t)$$

$$C(t) = \int g(t)\,\mu(t)\,dt + C$$

Thus $y(t) = C(t)\, e^{-\int p(t)\,dt} = \dfrac{C(t)}{\mu(t)} = \dfrac{1}{\mu(t)}\left[\int g(t)\,\mu(t)\,dt + C\right]$

2 methods → ~ integrating factor
→ ~ variation of parameters

Solve : $\begin{cases} ty' + 2y = 4t^2 \\ y(1) = 2 \end{cases}$

Step 1 : $ty' + 2y = 0$ ---- ①

$$\frac{1}{y}\frac{dy}{dt} = -\frac{2}{t}$$

$$\frac{1}{y}\frac{dy}{dt} = \frac{d}{dt}\ln|y|$$

$$\Rightarrow \ln|y(t)| = -2\int \frac{dt}{t}$$

$$= -2\ln t + C$$

$$= \ln\frac{1}{t^2} + C$$

① $\ln|y| + 2\ln t = C$

or $\ln|yt^2| = C$

$$y(t) = \frac{C}{t^2}$$

Step 2 ; try $y(t) = \frac{C(t)}{t^2}$

into $y' + \frac{2}{t}y = 4t$

$$y' = C'(t)\frac{1}{t^2} + C(t)(-2t^{-3})$$

$$= \frac{C'}{t^2} - \frac{2}{t}\frac{C(t)}{t^2}$$

$$y' + \frac{2}{t}y = \frac{C'(t)}{t^2} = 4t$$

$$\Rightarrow C'(t) = 4t^3 \Rightarrow C(t) = t^4 + C$$

$$y(t) = \frac{C(t)}{t^2} = \frac{t^4 + C}{t^2} = \frac{C}{t^2} + t^2$$

$$2 = y(1) = \frac{C}{1^2} + 1^2 = C + 1$$

$$\Rightarrow C = 1$$

$$y(t) = \frac{1}{t^2} + t^2, \quad t > 0$$